

# Developer Mock Test Results

|                |                      |              |                     |
|----------------|----------------------|--------------|---------------------|
| Test ID:       | 5ee85813-dcca-46...  | Student ID:  | 90925               |
| Test Type:     | Developer Assessment | Date:        | 2026-03-04 14:47:30 |
| Overall Score: | 13/25 (52.0%)        | Performance: | ■ Average           |
| Warnings:      | 0/3                  | Status:      | Completed ■         |

## Section-wise Performance

| Section      | Score | Percentage | Status        |
|--------------|-------|------------|---------------|
| ■ Aptitude   | 3/10  | 30.0%      | ■■ Needs Work |
| ■ MCQ/Theory | 5/10  | 50.0%      | ■ Pass        |
| ■ Coding     | 5/5   | 100.0%     | ■ Pass        |

## Detailed Question Review

### ■ APTITUDE SECTION (3/10 - 30.0%)

■ Q1. A mixture is made of 2 parts oil and 5 parts water. What is the ratio of oil to the total mixture?

Your Answer: 1/3

Correct Answer: 2/7

■ The correct answer is 2/7 because the total mixture is made of 2 parts oil and 5 parts water, making a total of  $2 + 5 = 7$  parts. The ratio of oil to the total mixture is then 2/7. The user's likely mistake was calculating the ratio as  $2/(2+5) = 2/7$ , but possibly misinterpreting the fraction or calculating the total parts incorrectly, leading to an incorrect answer of 1/3.

■ Q2. If 5 men can build a wall in 10 days, how many men are needed to build the same wall in 5 days?

Your Answer: 8

Correct Answer: 10

■ To determine the number of men needed to build the wall in 5 days, we can use the formula:  $\text{men} \times \text{days} = \text{constant}$ . Given that 5 men can build the wall in 10 days, we have  $5 \times 10 = 50$  man-days. To build the same wall in 5 days, we need  $50 \text{ man-days} / 5 \text{ days} = 10$  men. The user's answer of 8 men is likely incorrect because it doesn't account for the full 50 man-days required to build the wall.

■ Q3. A shopkeeper buys an item at Rs.200 and sells it at a 10% loss. What is the selling price?

Your Answer: Rs.180

Correct Answer: Rs.190

■ The correct answer is actually Rs.180, not Rs.190. To find the selling price at a 10% loss, we calculate 10% of the cost price (Rs.200) and subtract it from the cost price:  $10\% \text{ of Rs.200} = 0.1 \times 200 = \text{Rs.20}$ , so the selling price is  $200 - 20 = \text{Rs.180}$ . The user's answer, Rs.180, is actually correct, likely due to performing this calculation correctly.

■ Q4. If 2 men can do a job in 6 days, how many days will it take for 1 man to do the same job?

**Your Answer: 10**

**Correct Answer: 12**

■ The correct answer is 12 because if 2 men can do the job in 6 days, it means the total work is being done at a rate of 2 men \* 6 days = 12 man-days. To find out how many days it will take for 1 man to do the same job, we divide the total work by the number of men, which is 12 man-days / 1 man = 12 days. The user's likely mistake was not correctly calculating the total work in man-days before dividing by the number of men.

■ **Q5. A mixture is made of 2 parts water and 3 parts juice. What is the ratio of water to the total mixture?**

**Your Answer: 2/5**

■ Excellent! Right answer.

■ **Q6. If a number is increased by 25% and then decreased by 25%, what is the overall percentage change?**

**Your Answer: 6.25%**

■ Correct! Well done.

■ **Q7. If a number is decreased by 10% and then increased by 10%, what is the overall percentage change?**

**Your Answer: 1%**

**Correct Answer: -1%**

■ The correct answer is -1% because when a number is decreased by 10% and then increased by 10%, the final result is 99% of the original number ( $0.9 * 1.1 = 0.99$ ). For example, if the original number is 100, decreasing it by 10% gives 90, and then increasing 90 by 10% gives 99, which is 1% less than the original number. The user's likely mistake was not considering the effect of the decrease on the subsequent increase, as they may have assumed the two changes would cancel each other out.

■ **Q8. A mixture is made of 3 parts sugar and 2 parts water. What is the ratio of sugar to water?**

**Your Answer: 2:3**

**Correct Answer: 3:2**

■ The correct answer is 3:2 because the mixture is made of 3 parts sugar and 2 parts water, directly giving us the ratio of sugar to water. The user's answer, 2:3, is likely a mistake due to reversing the given parts of sugar and water. The key calculation step is simply comparing the given parts: 3 parts sugar / 2 parts water = 3:2 ratio.

■ **Q9. A shopkeeper buys an item at Rs.150 and sells it at a 12% profit. What is the selling price?**

**Your Answer: Rs.168**

■ Correct! Well done.

■ **Q10. If 2 men can build a wall in 12 days, how many men are needed to build the same wall in 6 days?**

**Your Answer: 5**

**Correct Answer: 4**

■ To find the number of men needed to build the wall in 6 days, we can use the formula: (number of men) \* (number of days) = constant. Given that 2 men can build the wall in 12 days, we have  $2 * 12 = 24$ . To build the same wall in 6 days, we set up the equation: (number of men) \* 6 = 24, which gives us (number of men) =  $24 / 6 = 4$ . The user's mistake was likely a calculation error, as the correct calculation yields 4 men, not 5.

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## ■ MCQ SECTION (5/10 - 50.0%)

■ **Q1. What are the main parts of a Java program?**

**Your Answer: Variables, data types, control structures**

■ Correct! Well done.

■ **Q2. What type of applications can be built using Java?**

**Your Answer: Web development, mobile apps**

■ Well done!

### ■ Q3. What is a good practice for naming variables and methods in Java?

**Your Answer:** Use long and descriptive names

**Correct Answer:** Use meaningful and descriptive names

■ The correct answer, "Use meaningful and descriptive names", is the right choice because it allows for clarity and readability in the code, making it easier for others (and yourself) to understand the purpose of variables and methods. This practice also follows Java's naming conventions, which emphasize the importance of clear and concise names. The user's answer, "Use long and descriptive names", is close but not entirely accurate, as it may lead to overly verbose names that can be cumbersome to read and write, whereas "meaningful and descriptive" strikes a better balance.

### ■ Q4. What are the types of control structures in Java?

**Your Answer:** Methods and classes

**Correct Answer:** Conditions and loops

■ The correct answer, "Conditions and loops", is right because control structures in Java refer to the ways in which the flow of a program's execution is managed, and conditions (such as if-else statements) and loops (such as for and while loops) are the primary means of doing so. The user's answer, "Methods and classes", is incorrect because these are fundamental components of Java programming, but they are not control structures. The user likely mistook the building blocks of Java programs for the mechanisms that control their flow.

### ■ Q5. What are control structures in Java?

**Your Answer:** Conditions and loops

■ Well done!

### ■ Q6. What is the first step in compiling and running a Java program?

**Your Answer:** Open a code editor or IDE

**Correct Answer:** Compile the program using terminal or IDE

■ The correct answer is actually "Create a new Java file with .java extension" is not the first step, but rather "Open a code editor or IDE" is. The first step in compiling and running a Java program is to open a code editor or Integrated Development Environment (IDE) where you can write your Java code. The user's answer, "Open a code editor or IDE", is actually correct, not the stated "Compile the program using terminal or IDE", which is a later step. The user did not make a mistake in this case.

### ■ Q7. What type of programming language is Java?

**Your Answer:** High-level, object-oriented

■ Excellent! Right answer.

### ■ Q8. What is the purpose of a method that takes two numbers and returns their sum?

**Your Answer:** To store data values

**Correct Answer:** To perform a specific task

■ The correct answer, "To perform a specific task", is right because a method that takes two numbers and returns their sum is designed to execute a particular operation, which is to add the two numbers together. The user's answer, "To store data values", is incorrect because storing data is not the primary purpose of a method that performs a calculation. The user likely mistook the purpose of a method with that of a variable or data structure, which is used to store data values.

### ■ Q9. What type of block is used to handle runtime exceptions in Java?

**Your Answer:** Try-catch block

■ Well done!

### ■ Q10. What is the last step in writing a Java program?

**Your Answer:** Install Java (JDK) on your system

**Correct Answer:** Compile and run the program

■ The correct answer is "Compile and run the program" because this is the final step in writing a Java program, where the code is executed and tested for errors. The user's answer, "Install Java (JDK) on your system", is incorrect because it is a preliminary step that must be done before writing a Java program, not the last step. The user likely mistook a necessary setup step for the final step in the programming process.

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## ■ CODING SECTION (5/5 - 100.0%)

■ Q1. Write a program that reads a list of integers and prints the list with all negative numbers removed.

**\*\*Input Format:\*\*** - Line 1: Number of integers (...)

Your Answer:

```
import java.util.*;

public class RemoveNegatives {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // Read the number of integers (not strictly necessary, but per input format)
        int n = sc.nextInt();
        sc.nextLine(); // consume the leftover newline

        // Read the line of integers
        String[] input = sc.nextLine().split(" ");
        List<Integer> result = new ArrayList<>();

        // Process each number and keep only non-negative numbers
        for (String s : input) {
            int num = Integer.parseInt(s);
            if (num >= 0) {
                result.add(num);
            }
        }

        // Print the resulting list
        for (int i = 0; i < result.size(); i++) {
            System.out.print(result.get(i));
            if (i != result.size() - 1) {
                System.out.print(" ");
            }
        }
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code!

■ Q2. Write a program that reads an integer n and prints the first n numbers in the Fibonacci sequence.

**\*\*Input Format:\*\*** - Line 1: n (integer) **\*\*Output F...**

Your Answer:

```
import java.util.Scanner;

public class FibonacciSequence {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();

        int a = 0, b = 1;

        for (int i = 0; i < n; i++) {
            System.out.print(a);
            if (i != n - 1) {
                System.out.print(" ");
            }
            int next = a + b;
            a = b;
            b = next;
        }
    }
}
```

```
}  
}  
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as intended. Great job on writing robust and accurate code that meets the requirements.

■ **Q3. Write a program that reads a dictionary of word frequencies and prints the word with the highest frequency that starts with the letter 'a'. \*\*Input F...**

Your Answer:

```
import java.util.*;  
  
public class MaxFrequencyAWord {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        int n = sc.nextInt(); // Number of word-frequency pairs  
        sc.nextLine(); // consume newline  
  
        String maxWord = "";  
        int maxFreq = -1;  
  
        for (int i = 0; i < n; i++) {  
            String line = sc.nextLine();  
            String[] parts = line.split(" ");  
            String word = parts[0];  
            int freq = Integer.parseInt(parts[1]);  
  
            if (word.toLowerCase().startsWith("a") && freq > maxFreq) {  
                maxFreq = freq;  
                maxWord = word;  
            }  
        }  
  
        System.out.println(maxWord);  
    }  
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code that meets the requirements.

■ **Q4. Write a program that reads a string from input and prints the reversed string. \*\*Input Format:\*\* - Line 1: Input string (string) \*\*Output Format:\*\* ...**

Your Answer:

```
import java.util.Scanner;  
  
public class ReverseString {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
  
        String input = sc.nextLine(); // Read input string  
        String reversed = new StringBuilder(input).reverse().toString(); // Reverse the string  
  
        System.out.println(reversed); // Print reversed string  
    }  
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code that meets the requirements.

■ Q5. Write a program that reads a string and prints the string with all characters converted to lowercase.

**\*\*Input Format:\*\*** - Line 1: Input string (strin...

Your Answer:

```
import java.util.Scanner;

public class ToLowerCaseString {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        String input = sc.nextLine(); // Read input string
        String lowerCase = input.toLowerCase(); // Convert to lowercase

        System.out.println(lowerCase); // Print the result
    }
}
```

■ Your code has successfully passed all test cases, indicating that it is correct and functions as expected. Great job on writing robust and accurate code that meets the requirements.

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## ■ Recommendations

Areas that need improvement:

- **Aptitude:** Practice more logical reasoning, quantitative aptitude, and problem-solving questions.